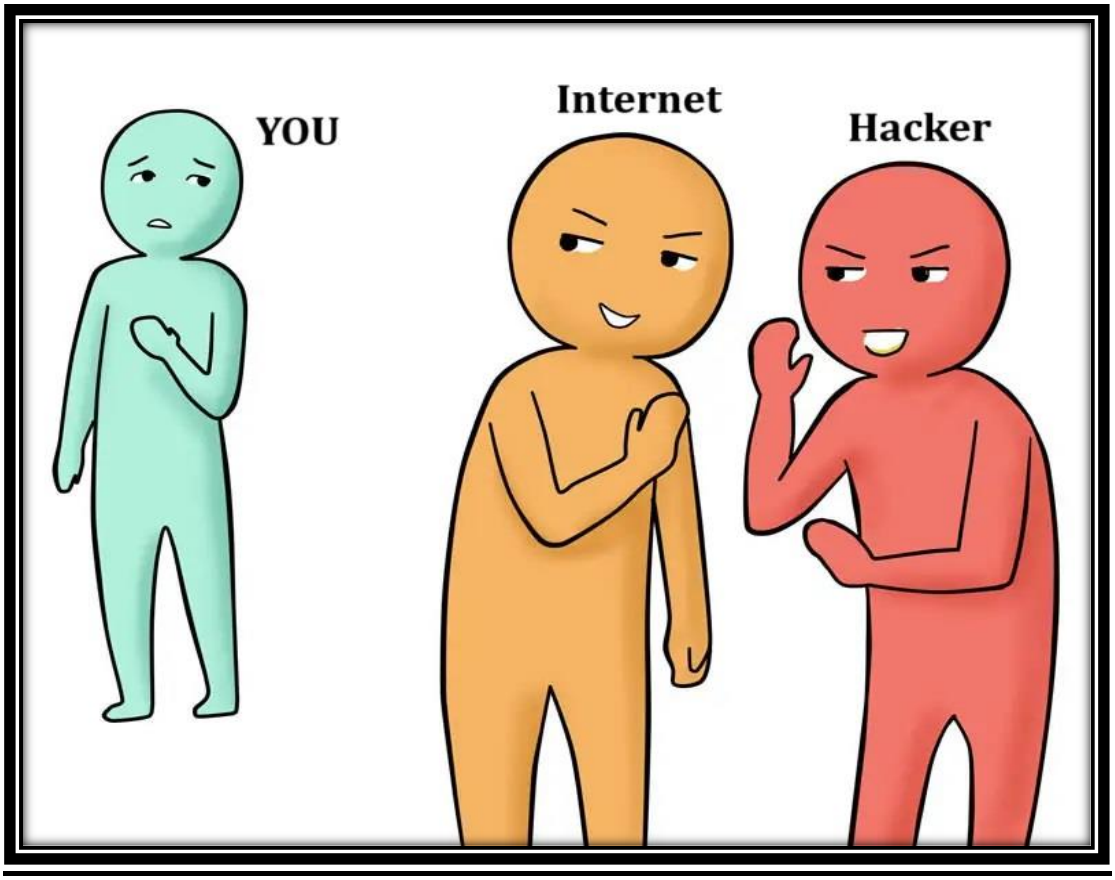
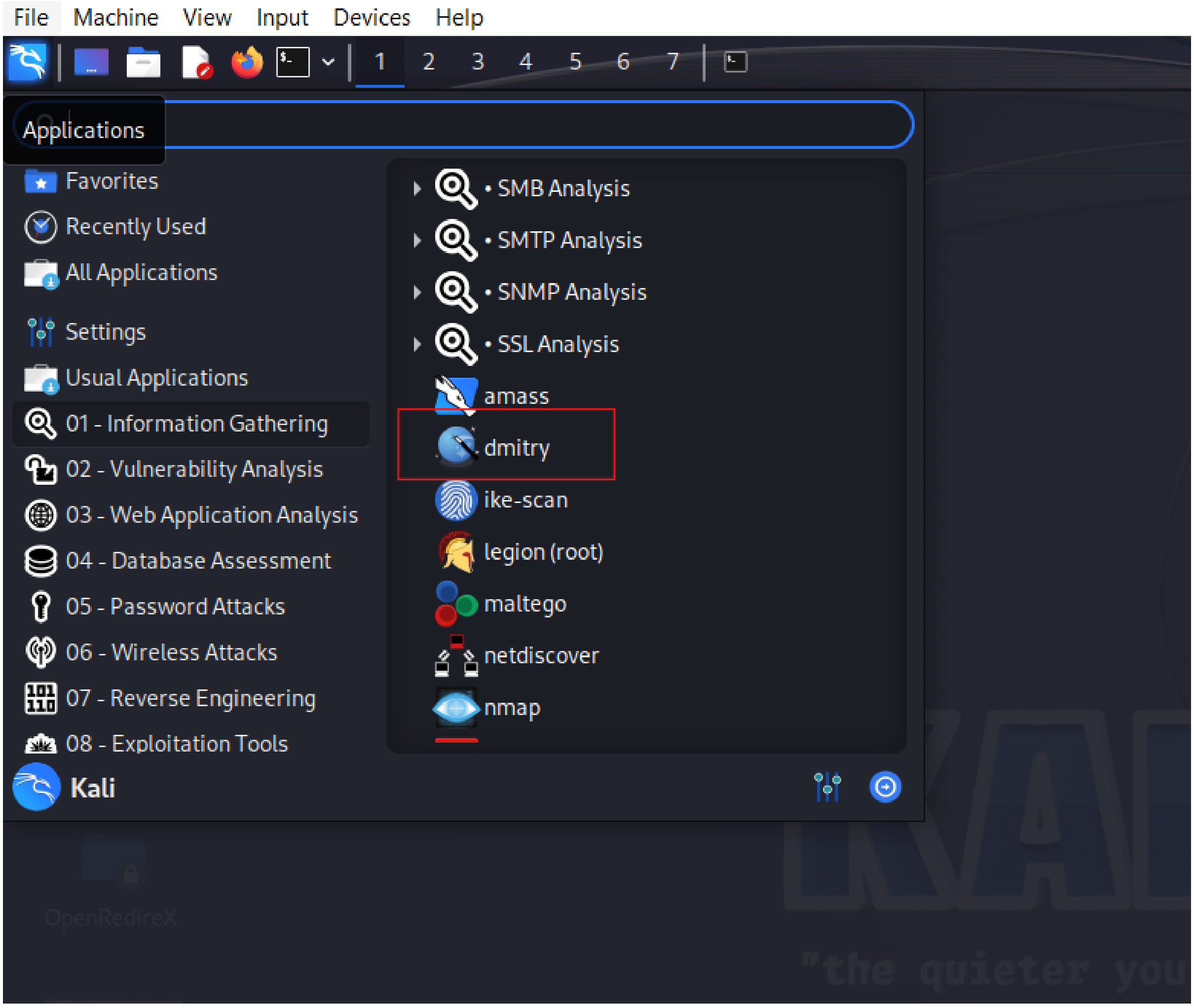
Dmitry
Information Gathering Tool



What is Information Gathering

The most important and initial phase in penetration testing for an ethical hacker is information gathering. Information about the target domain or network is gathered throughout this phase.
During the pen-testing process, a variety of information categories, namely Submain Enumeration, Port Information, DNS Information, Email Intel, Technology Stack, etc., can be helpful.
To perform this task, there are several Open-Source Tools available, one of them is the in-built tool in Kali called Dmitry.

Dmitry



Installation:

Dmitry is pre-installed and directly accessible in Kali.

Or

one can clone and download the Dmitry tool from GitHub. Clone the tool using the following command

command: git clone https://github.com/jaygreig86/dmitry.git

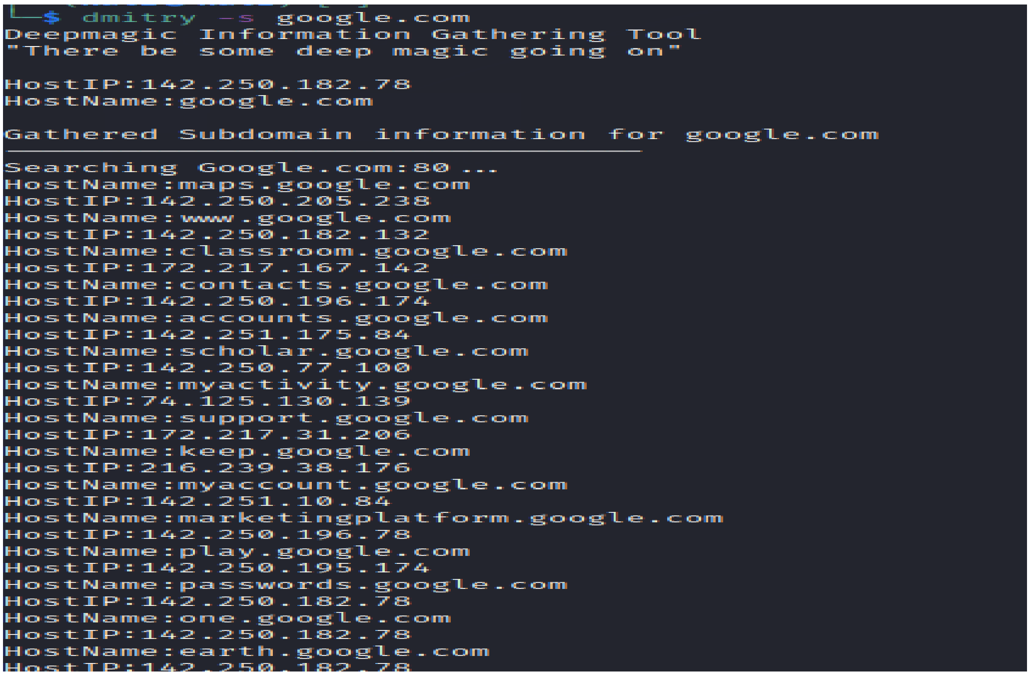
Usage:

```
File  Actions  Edit  View  Help
$ dmitry
Deepmagic Information Gathering Tool
"There be some deep magic going on"

Usage: dmitry [-winsepfb] [-t 0-9] [-o %host.txt] host
  -o     Save output to %host.txt or to file specified by -o file
  -i     Perform a whois lookup on the IP address of a host
  -w     Perform a whois lookup on the domain name of a host
  -n     Retrieve Netcraft.com information on a host
  -s     Perform a search for possible subdomains
  -e     Perform a search for possible email addresses
  -p     Perform a TCP port scan on a host
* -f     Perform a TCP port scan on a host showing output reporting filtered ports
* -b     Read in the banner received from the scanned port
* -t 0-9 Set the TTL in seconds when scanning a TCP port ( Default 2 )
*Requires the -p flagged to be passed
```

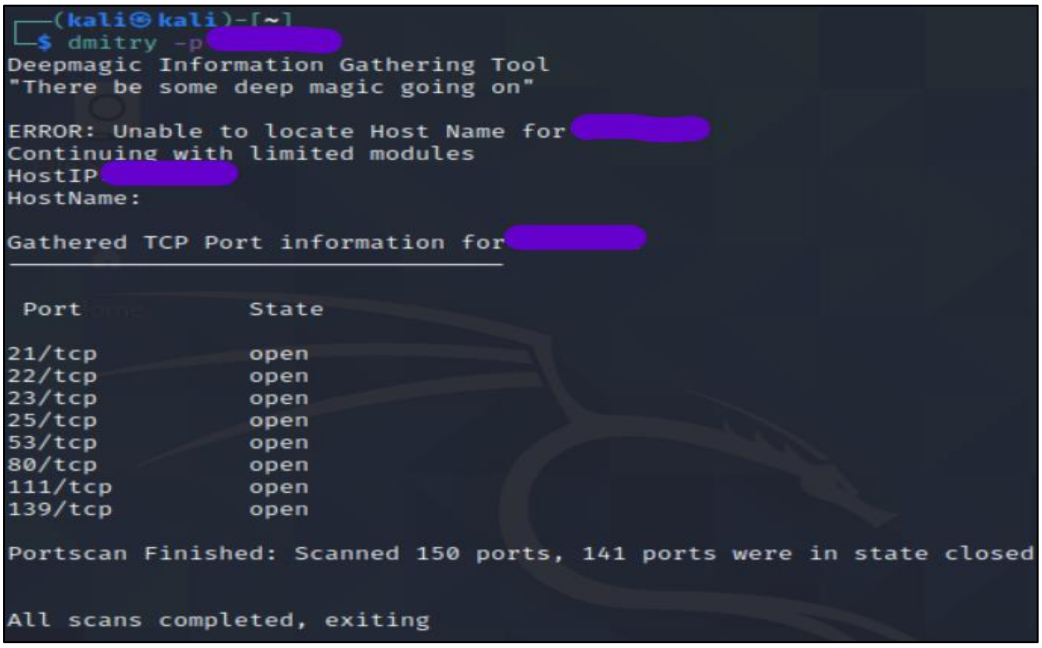
Flags	Uses of flags
1. -o	Allows the user to specify a location to write the output of the application to.
2. -i	This flag Performs a whois lookup on the IP address of the target.
3. -w	This flag Performs a whois lookup on the domain name of the host.
4. -n	This flag Retrieves all available Netcraft information for a given target.
5. -s	This flag is used to search all subdomains of the target.
6. -e	This flag is used to search all emails associated with the target.
7. -p	This flag performs TCP port scanning and find open ports of the target

1) Subdomain:



The tool can be used to gather the possible subdomains of a target. A subdomain can further be utilized to identify and exploit login pages, admin panels, subdomain takeover attacks etc.

2) Open Ports:



The tool can be used to discover the open ports of the target which is one of the essential parts of reconnaissance. Non-essential ports being open could lead to jeopardizing the whole network which might further damage the reputation of a firm

3) Whois:

Whois will provide all the details required to determine who is the domain's owner, where it is registered, what nameservers it uses when it was registered, and when it might expire, nameservers etc.

One can check the whois information with both hostname and IP.

```
┌──(kali㉿kali)-[~]
└─$ dmitry -w tryhackme.com
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HostIP:104.22.54.228
HostName:tryhackme.com

Gathered Inic-whois information for tryhackme.com
─────────────────────────────────
   Domain Name: TRYHACKME.COM
   Registry Domain ID: 2282723194_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.namecheap.com
   Registrar URL: http://www.namecheap.com
   Updated Date: 2021-05-01T19:43:23Z
   Creation Date: 2018-07-05T19:46:15Z
   Registry Expiry Date: 2027-07-05T19:46:15Z
   Registrar: NameCheap, Inc.
   Registrar IANA ID: 1068
   Registrar Abuse Contact Email: abuse@namecheap.com
   Registrar Abuse Contact Phone: +1.6613102107
bited https://icann.org/epp#clientTransferProhibited
   Name Server: KIP.NS.CLOUDFLARE.COM
   Name Server: UMA.NS.CLOUDFLARE.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2024-09-22T12:13:18Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
ation in
 the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
cessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
```

4) Netcraft information of a host:

To obtain the required data, including the operating system, web server, web host, hosting provider, and so on, Dmitry Tool can be utilized in cooperation with the Netcraft service. This information can be exploited if any data is found to be outdated or vulnerable such as vulnerable server version etc.

5) Email Address:

```
└─$ dmitry -e google.com
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"There be some deep magic going on"

HostIP:142.250.77.142
HostName:google.com

Gathered E-Mail information for google.com
---------------------------------
Searching Google.com:80 ...
abc@google.com
nobody@google.com
postmaster@aspmx.l.google.com
ter@aspmx.l.google.com
info@google.com
earthdok@google.com
Searching Altavista.com:80 ...
```

Dmitry can be used to obtain the email address of a target system which can further be exploited to perform malicious acts like spear phishing attack, or any other social engineering attack.

Conclusion:

Despite its modest size, Dmitry has always been a really interesting tool. It is simple to use and ideal for someone who has just started learning about CLI, Linux, and tools and techniques used in cyber security.

Gathering information is the initial stage in both pen testing and if someone is interested in bug bounty as well. Dmitry is quite handy and one of the best tools for learners who just stepped into the realm of cybersecurity.

References:

Please refer to the below links for a detailed explanation:

- https://www.geeksforgeeks.org/dmitry-passive-information-gathering-tool-in-kali-linux/
- https://www.youtube.com/watch?v=8SnOexSoNdk&ab_channel=InnoTechTips
- https://techyrick.com/dmitry-what-is-full-tutorial-from-scratch/